

Evaluating Zero-Shot CLIPSeg Models on Indoor Object Segmentation

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Abstract—This paper explores the performance of CLIPSeg, a vision-language model, for semantic segmentation on indoor scenes from the ML-Hypersim dataset. We evaluated three pre-trained CLIPSeg variants in a zero-shot setting on a small labeled subset of 4 object classes. The goal was to apply different variations of zero-shot models and compare the accuracy while tweaking the given prompts and thresholds. The results show that changing the threshold affects the models differently. After computing per-class metrics, the more refined and in-depth models generally perform better across the selected classes.

Index Terms—ClipSeg, Image-Segmentation, zero-shot-segmentation, prompt-based model, vision-language model

I. INTRODUCTION

Hypersim is a photorealistic synthetic dataset for holistic indoor scene understanding. The dataset was created by leveraging a large repository of synthetic scenes created by professional artists. Overall it includes a 74000 images with per-pixel labels and corresponding ground truth geometry. In this project, we focus on applying and evaluating CLIPSeg, a vision-language model that can perform segmentation using natural language prompts. Our main goal is to understand how well different pre-trained versions of CLIPSeg perform out-of-the-box on ML-Hypersim data.

II. METHODS

A. CLIPseg

CLIPSeg is a vision-language model for image segmentation, combining the capabilities of the CLIP architecture with a segmentation decoder. Unlike traditional segmentation models that rely solely on pixel-wise classification, CLIPSeg incorporates natural language prompts as conditioning input, enabling zero-shot segmentation based on textual queries (e.g., “a bed” or “a window”) [1] [3] [4] .

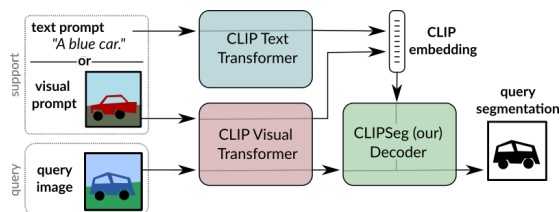


Figure 1 – CLIPSeg Architecture

The model architecture consists of two main parts:

- 1) **CLIP Backbone** – A frozen CLIP encoder (typically a ViT or ResNet) extracts dense visual features from the input image, while a text encoder transforms natural language prompts into a shared embedding space.
- 2) **Segmentation Decoder** – A lightweight decoder receives the fused visual and textual embeddings and produces pixel-wise segmentation masks corresponding to the input query.

CLIPSeg supports both zero-shot and fine-tuned usage modes:

- **Zero-shot:** The model segments objects using only the provided text prompt, without additional training on the dataset.
- **Fine-tuned:** A small dataset of image-mask pairs is used to optimize the decoder, improving accuracy for specific domains or tasks.

III. DATA

The dataset [2] contains indoor scene renderings paired with semantic segmentation masks. The original dataset is structured using .hdf5 files for both color images and corresponding semantic labels.

From the full dataset, we selected a subset of 4 semantic classes based on relevance and visual clarity. Each class was mapped to a simple text prompt, such as “a wall” or “a chair”, which was later used as input for zero-shot. The class ID-to-prompt mapping allowed us to evaluate the model’s ability to associate visual regions with language.

The selected classes were “Chair”, “Table”, “Floor”, and “Bed”. Chair and Table were included because they are similar, which could make them more challenging to separate.

Each image was resized to 352×352 resolution, and binary masks were created for each class label, transforming the multi-class task into multiple binary segmentation problems, which is how the CLIPSeg model interprets prompt-based segmentation.

To manage computational complexity, we only used 2505 samples for model evaluation.

IV. RESULTS

A. Model CIDAS/clipseg-rd64-refined

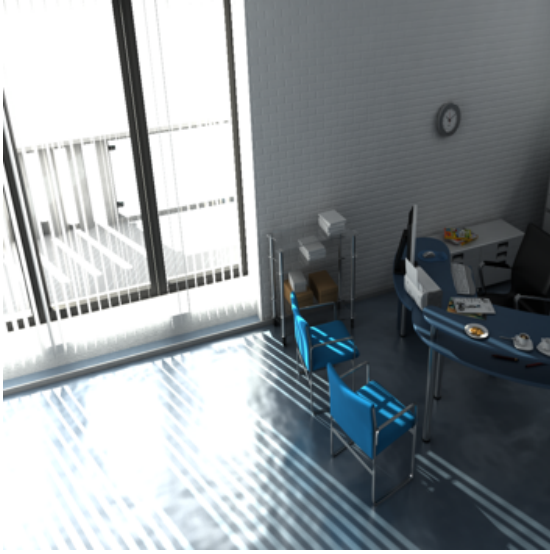
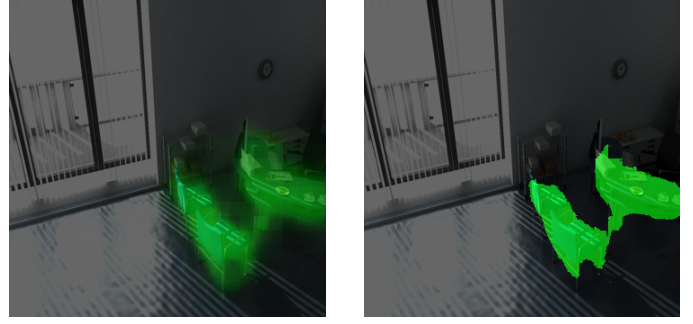


Figure 2 – Original Image



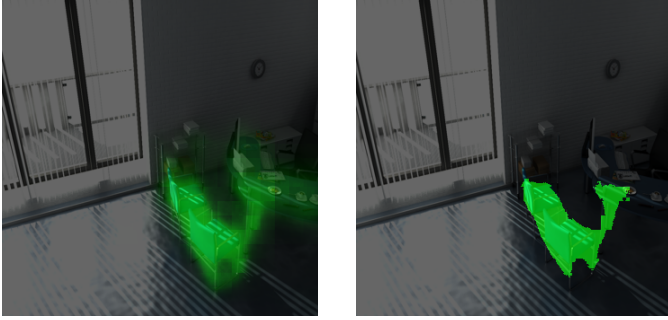
Prompt "Chair and Table"

Threshold 0.4

Figure 5 – Output of the model **CIDAS/clipseg-rd64-refined**

The model was tested with the prompt "Chair". Initially, without thresholding, it showed confusion between *chair* and *table*. After applying a **0.6 threshold**, it correctly identified **blue chairs**, but struggled with *black chair*. Even with the prompt "Black Chair", the model failed to detect them. However, the combined prompt "Chair and Table" yielded accurate segmentation for both objects. For these cases, the best results were obtained using a threshold of **0.6** for "Black Chair" and **0.4** for "Chair and Table".

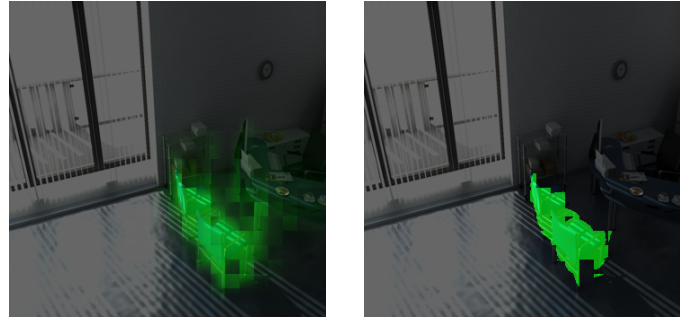
B. Model CIDAS/clipseg-rd64



Prompt "Chair"

Threshold 0.6

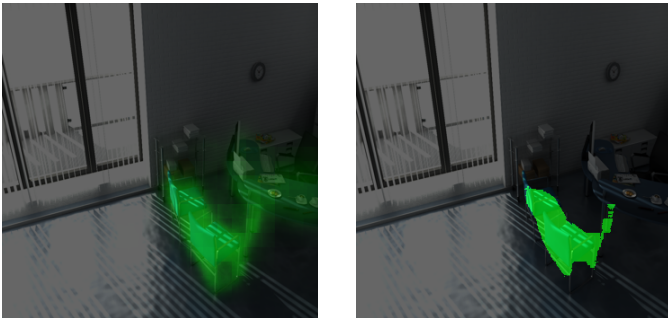
Figure 3 – Output of the model **CIDAS/clipseg-rd64-refined**



Prompt "Chair"

Threshold 0.6

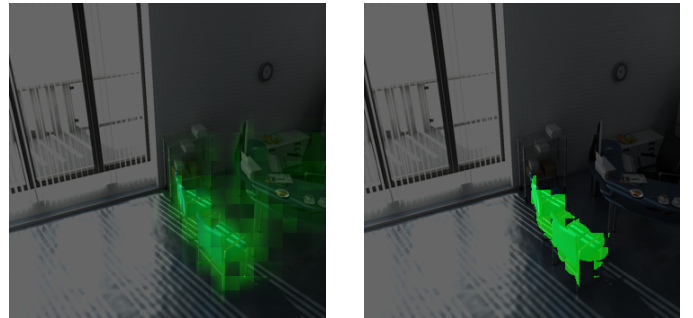
Figure 6 – Output of the model **CIDAS/clipseg-rd64**



Prompt "Black Chair"

Threshold 0.6

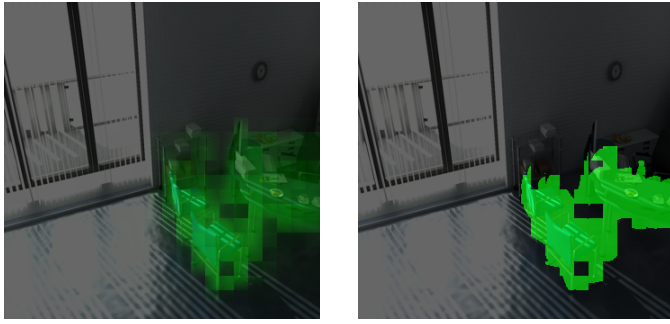
Figure 4 – Output of the model **CIDAS/clipseg-rd64-refined**



Prompt "Black Chair"

Threshold 0.7

Figure 7 – Output of the model **CIDAS/clipseg-rd64**



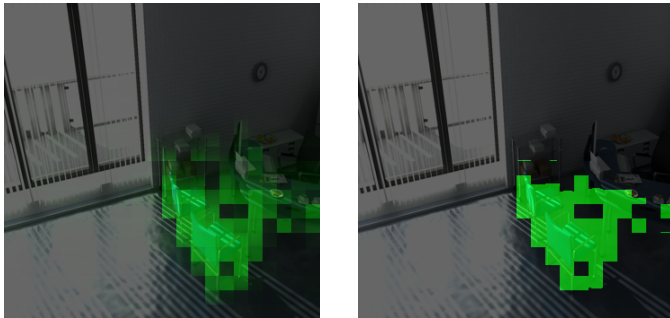
Prompt "Chair and Table"

Threshold 0.35

Figure 8 – Output of the model **CIDAS/clipseg-rd64**

When evaluating the **CIDAS/clipseg-rd64** model, we observed that model reduced confusion with tables. However, the resulting segmentation masks appeared *noisier* compared to those from the refined model. The model consistently failed to detect the *black chair*, even with color-specific prompts. When using the prompt "*Chair and Table*", the model struggled with edge definition and produced more noise in the predictions. For these cases, the best results were obtained using a threshold of **0.6** for "*Black Chair*" and **0.35** for "*Chair and Table*".

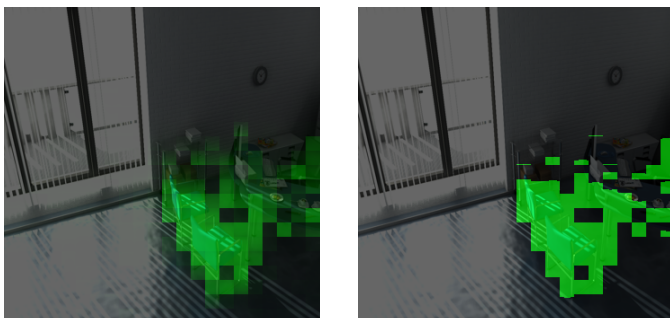
C. Model CIDAS/clipseg-rd16



Prompt "Chair"

Threshold 0.4

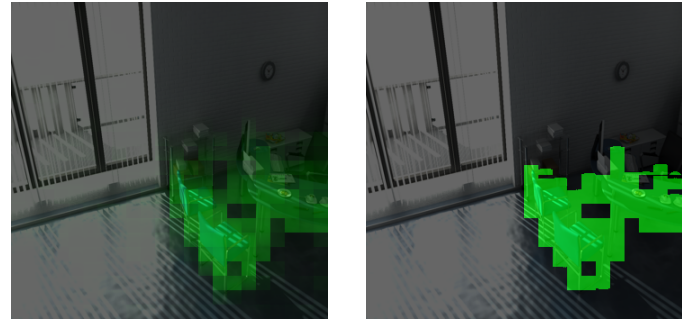
Figure 9 – Output of the model **CIDAS/clipseg-rd16**



Prompt "Black Chair"

Threshold 0.4

Figure 10 – Output of the model **CIDAS/clipseg-rd16**



Prompt "Chair and Table"

Threshold 0.3

Figure 11 – Output of the model **CIDAS/clipseg-rd16**

The **CIDAS/clipseg-rd16** model performed noticeably worse than the other two variants. While it was able to roughly localize objects such as *chairs* and *tables*, the output masks appeared more *blocky and rectangular*, lacking the smoothness and precision seen in the other models. The segmentation around object edges was particularly coarse, with boundaries often captured by large rectangular shapes, resulting in a noisier prediction.

We also compared the models by evaluating per-class metrics:

Model	Prompt	F1	Accuracy	Sensitivity	Specificity
rd64-refined	the floor	0.721299	0.938724	0.698056	0.974041
rd64-refined	a chair	0.457096	0.949868	0.560720	0.967503
rd64-refined	a bed	0.537657	0.938833	0.530287	0.974754
rd64-refined	a table	0.283398	0.956490	0.324818	0.971335
rd64	the floor	0.717041	0.935251	0.678875	0.972882
rd64	a chair	0.409269	0.948065	0.474542	0.971384
rd64	a bed	0.507191	0.939282	0.509004	0.973837
rd64	a table	0.273899	0.953391	0.366174	0.968542
rd16	the floor	0.704214	0.931478	0.663723	0.971526
rd16	a chair	0.365508	0.944872	0.430006	0.968466
rd16	a bed	0.553340	0.941451	0.595965	0.955857
rd16	a table	0.222279	0.947818	0.323232	0.962789

Table I – Zero-shot evaluation of CLIPSeg variants

Looking at the results, the performance across different models is quite close, but the more refined and deep models do seem to be a bit more consistent. This is especially noticeable when it comes to classes that are somewhat similar, like chairs and tables, where the refined models seem to handle the differences a little better.

Code

REFERENCES

- [1] Till Blaha, Robin Rombach, Patrick Esser, and Björn Ommer. CLIPSeg - image segmentation using text and image prompts. <https://github.com/timofjl/clipseg>, 2022. Accessed: 2025-05-25.
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- [4] Lihu Pan, Yunting Yang, Zhengkui Wang, Rui Zhang, Wen Shan, and Jiashu Li. Image segmentation with vision-language models. In *Proceedings of the 2023 7th International Conference on Computer Science and Artificial Intelligence*, pages 233–238, 2023.